A versatile FMCW Radar System Simulator for Millimeter-Wave Applications

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***Abstract*— For the successful design of low-cost and high-performance radar systems accurate and efficient system simulation is a key requirement. In this paper we present a new versatile simulation environment for frequency-modulated continuous-wave radar systems. Besides common hardware simulation it covers integrated system simulation and concept analysis from signal synthesis to baseband. It includes a flexible scenario generator, accurate noise modeling, and efficiently delivers simulation data for development and testing of signal processing algorithms. A comparison of simulations and measurement results for an integrated 77-GHz radar prototype shows the capabilities of the simulator on two different scenarios.**

I. INTRODUCTION

Wireless distance, angle, and velocity sensing by means of radar measurements is an active research topic. In contrast to most military applications, which benefit from substantial funding, in industrial and especially in automotive applications the system costs must be kept at a minimum for economical success. The costs for system design, prototyping, and redesign account for a significant part of the total development expenses, therefore accurate computer simulation and concept analysis is a valuable design aid. On the one hand the system simulation allows the designer to investigate different conceptual approaches for their strong points and shortcomings before prototyping, which otherwise is time-consuming and costly. On the other hand, it prevents over-specified system designs and allows to meet an accurate trade-off between the selected components, which results in a balanced and efficient system concept, from a technical as well as economical point of view.

Of course, basic radar simulation is not a new topic. The traditional simulation of radio frequency (RF) hardware components is achieved by many commercially available software packages, nevertheless this is seldom sufficient if the performance of the complete system should be retrieved. Especially in frequency-modulated continuous-wave (FMCW) radar systems [1], which are investigated in this contribution, a good understanding of additional parameters, such as the frequency synthesis, e.g. frequency modulations, sweep linearity, phase noise, etc., and the baseband interface is essential for optimization of the system's performance. These aspects are hardly treated jointly by existing solutions. Furthermore it is advisable to account for the capabilities and limitations of the applied digital signal processing (DSP) techniques, which requires an interdisciplinary simulation approach. Regarding these aspects, we present a novel simulation environment that is applicable to hardware design as well as algorithm development, and furthermore efficiently generates accurate and reliable data sets for algorithm testing and performance analysis.

II. SIMULATOR STRUCTURE

In the following, we will shortly introduce the modular structure of the simulator, which is completely implemented in MATLAB®. A comparison of simulation and measurement results for some specific test cases is given in the final part.

A. Baseband System Simulation

Simulation time is crucial, especially if a large number of data sets must be generated for the statistical analysis of DSP algorithms. Therefore the simulator uses an efficient baseband simulation technique, which calculates the system states at a time grid defined by the simulation clock f_{sim}. This clock is specified by the intermediate frequency (IF) bandwidth B_{IF} of the radar system, and can be significantly higher than the desired output data rate f_{S}. This approach ensures that aliasing of noise and frequency components within the total IF bandwidth is correctly included. The final baseband data rate is achieved by a decimation process.

B. Simulator Block Schematic

Fig. 1 illustrates the main building blocks of the FMCW radar simulator.

1) Segmented Sweep Generation: At the input the transmit (TX) frequency course $f_{\mathrm{TX}}[n]$, with n the simulation sample index, is specified by piecewise linear frequency sweep segments, which are defined by the vertices of the frequency values. This approach does not only allow the flexible generation of up-down chirps at user-selectable ramp slopes, as used e.g. for range-Doppler processing, but is capable of modeling almost any arbitrary nonlinear frequency course by fine segmentation of the vertices.

2) *Frequency to Phase Conversion:* The transmit frequency course is converted to phase data $\varphi_{\mathrm{TX}}[n]$ by symbolic integration of the sweep segments. It is very important to accurately identify the sweep segment limits, since the corner frequencies of the TX frequency specification need not necessarily coincide with the temporal simulation grid, which otherwise would lead to erroneous phase offsets.

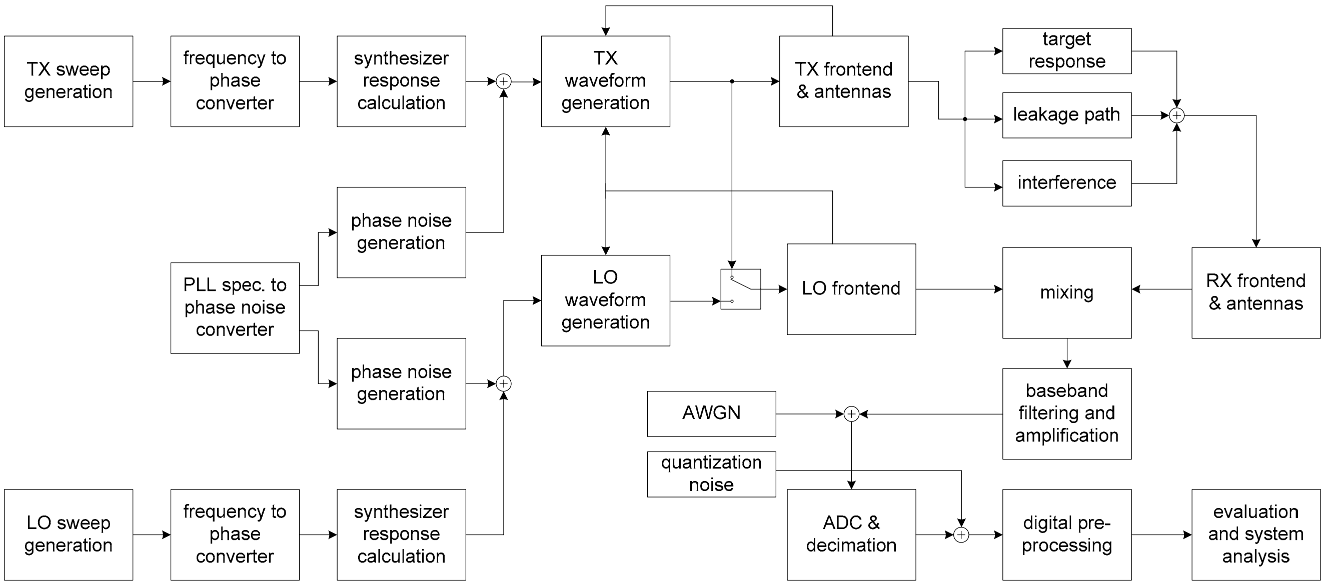

Fig. 1 Block schematic of the FMCW radar simulation environment, which is completely implemented in MATLAB®.

3) *Synthesizer Response Calculation:* In practical radar systems the dynamic performance of the frequency synthesizer is limited. Especially in phase-locked-loop (PLL) stabilized systems the loop filter limits the maximum achievable rate of change of the frequency output. This results in the typically observed overshoots and settling effects at sweep segment limits. To cover these effects, the simulator implements a linear PLL model for the phase domain. Furthermore, the synthesizer's phase noise specification is derived from the PLL model, using the noise characteristics of the PLL's reference oscillator, voltage controlled oscillator (VCO), phase-frequency-detector (PFD), and frequency divider.

4) Noise Simulation: The three main noise sources covered in the simulation are thermal input and component noise, quantization noise (QN) at the analog to digital converter (ADC) and phase noise (PN) introduced by the frequency sources. The available thermal noise power is obtained from the frontend simulation by integration oft the calculated noise power density on the effective noise bandwidth of the receiver. It is superposed on the IF data as an additive white Gaussian noise (AWGN) process. Quantization noise is defined by the effective signal-to-noise ratio (SNR) of the ADC and the sampling parameters as well as the ADC's full scale voltage. It is added as a white noise process, which is a good approximation in almost any practical radar application [2]. Finally, the phase noise contribution $\varphi_{\mathrm{PN}}[n]$ is calculated from the single-sideband phase noise specification of the frequency synthesizer in the frequency domain. It is added to the deterministic TX phase course before the noisy transmit signal is calculated.

5) *TX / LO Waveform Generation:* Using the deterministic TX phase course and a specific realization of the phase noise process the complex TX signal is computed to $s_{\mathrm{TX}}[n] = A_{\mathrm{TX}}[n]\exp(\mathrm{j}(\varphi_{\mathrm{TX}}[n] + \varphi_{\mathrm{PN}}[n]))$, with $A_{\mathrm{TX}}[n]$ the signal's amplitude. Optionally it is possible to account for external phase and amplitude modulation as well as phase and amplitude distortions obtained from the frontend simulation. Depending on the type of simulated receiver, the same or a different (and thus uncorrelated) phase noise realization is used for generating the mixer's local oscillator (LO) signal. This allows to model incoherent radar systems, which use a separate oscillator for TX and LO signal, and show significantly increased phase noise sensitivity [3] than conventional coherent FMCW units.

6) *RF Frontend Simulation and Antennas:* Besides the conventional frontend characteristics, such as cascaded gain, noise figure, electrical length, and compression analysis, the frontend module calculates frequency dependent amplitude and phase modulation properties, as well as leakage signals via s-parameter models. Special focus is set on the TX antenna's mismatch, and the limited isolation of TX and RX path, which in most applications represent the dominating leakage sources. The component specifications used for the frontend simulation are either derived from component simulations, measurement data, or data sheets. Furthermore co-simulation with external hardware simulators is supported.

7) Target Scenario Simulation: The individual target returns are calculated in a multi-target three-dimensional scenario simulator. It supports static, constant velocity, as well as constantly accelerated target models in any spatial direction, which allows to accurately model moving targets. Each individual target is parameterized by an initial position, velocity, and acceleration vector, as well as an angle dependent radar cross section (RCS), and angular velocity along the jaw axis. For the simulation of antenna arrays, a user selectable number of antennas is placed at defined positions with respect to their phase centers. Each antenna is characterized by its orientation and measured or simulated beam pattern. Furthermore the operational mode (TX or RX) over time can be specified. In every simulation time step, the scenario simulator identifies the round-trip delay-time (RTDT) of all signal paths that are currently involved. For this purpose it calculates the intersections of the moving targets and the propagating radiated electromagnetic waves.

8) *Mixer / RX Signal Simulation:* With knowledge of the – possibly varying – RTDTs of the simulated targets the phase courses $\varphi_{\mathrm{RX},i}[n]$ of the signals arriving at the receiver input are computed. Therefore the deterministic part of the TX phase is recalculated with consideration of the time shift caused by the actual RTDT. After conversion to the time domain, all target responses are superposed. This overlayed RX signal is fed to the mixer model, which currently implements ideal signal multiplication, but can be extended to nonlinear polynomial models or trained neuronal networks.

9) *Baseband Simulation:* In the baseband stage, the impact of IF filters and amplifiers on the mixer's IF output is simulated, using the component's transfer functions and noise contributions. As already stated, the IF filters set the final noise bandwidth of the total system. Therefore, if the IF bandwidth is larger than the ADC's Nyquist limit, noise and signal components located beyond are aliased to the first Nyquist zone. This increases the observed noise level and potentially generates unexpected spurs within the IF frequency range. The aliasing process is implemented by signal decimation from the simulation time step – which covers the whole IF bandwidth – to the finally desired sampling rate.

The ADC is specified with respect to its number of bits, full-scale input voltage, input resistance, and effective SNR, which defines the available quantization noise power. Furthermore, the transfer function of the ADC's analog input stage can be specified, which is useful for sub-sampling applications.

10) *Data Analysis and Signal Processing:* In the final processing stage, the simulated data is analyzed from a conceptual and signal processing point of view. From the achieved SNR valuable conclusions for a possible optimization of the system setup can be drawn by identification of the main bottle-necks. Moreover, the statistical analysis of the data allows to predict target resolvability as well as bias and variance of measurement parameters such as target range, position, angle, and velocity as a function of the system parameters. Furthermore the influence of different modulation techniques in the amplitude, phase, and frequency domain can easily be investigated.

III. Simulation and Measurement Results

To verify the accuracy of the simulator output, we compared simulation and measurement results of a state-of-the-art integrated 77-GHz FMCW radar prototype [4] based on a Silicon-Germanium (SiGe) chipset from Infineon. It consists of a 77-GHz VCO [5], a 77-GHz Gilbert-cell mixer, and a 19-GHz down-converter [6] for generating the LO signal. The VCO is stabilized in an offset-loop PLL, as shown in Fig. 2. To achieve agile frequency modulation, a direct digital synthesizer (DDS) drives the PLL's reference input. Two 4x4 patch arrays serve as transmit and receive antennas.

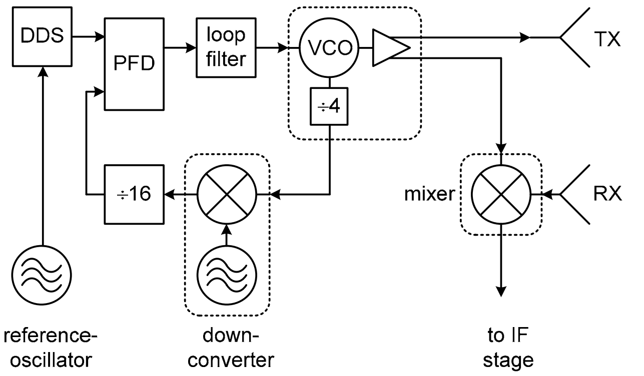

Fig. 2 Block schematic of the integrated 77-GHz FMCW radar prototype with separate transmit and receive antennas.

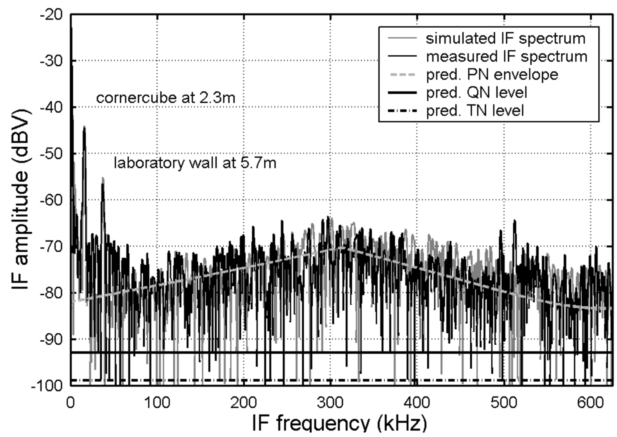

Fig. 3 Comparison of simulation and measurement results in a laboratory setup, additionally showing the predicted phase noise, quantization noise, and thermal noise levels.

A. Noise Simulation Results

Fig. 3 illustrates the comparison of simulation and measurement in a static scenario. The radar target is a corner cube with a RCS of –4 dBm^2 located at a distance of 2.3 m in the radar's main beam direction. A linear frequency sweep with bandwidth $B_{\mathrm{sw}} = 1$ GHz, starting at $f_0 = 75.5$ GHz and a sweep duration of $T_{\mathrm{sw}} = 1.012$ ms has been used. A 12-bit ADC, providing an effective SNR of 65.2 dB, samples the IF signal at a rate of $f_{\mathrm{S}} = 1.25$ MHz. The frontend radiates 25 dBm of transmit power and exhibits a total receiver gain of 44 dB. The noise figure of the RF receiver is relatively high at 16 dB, due to the missing low noise amplification stage prior the active mixer. The IF stage provides a gain of 24 dB and a first- order anti-aliasing filter with a bandwidth of 1.25 MHz. As can clearly be seen, the noise floor in the shown scenario is limited by the phase noise, which reproduces at the strong DC component that originates from the limited isolation between TX and RX antenna. Although the phase noise is strongly suppressed by correlation effects for low RTDTs, see eg. [7], the typical noise ear at the PLL's corner frequency of 2.3 MHz can clearly be identified. It aliases into the baseband frequency range due to the limited suppression of the anti-aliasing filter. The predicted phase noise envelope, as well as the synthetically simulated noise realization and the target response levels show good match to the measured result. Additionally, the predicted thermal and quantization noise levels are indicated.

B. Pass-By Scenario Simulation Results

In a second measurement set shown in Fig. 4, the radar unit is mounted on a computer-controlled linear rail and passes a static target, which is formed by a corner cube reflector. Thus the horizontal aspect angle φ of the target significantly changes during the measurement. In this case the received power strongly depends on the beam pattern of the radar module and the angle-dependent reflection characteristics of the target.

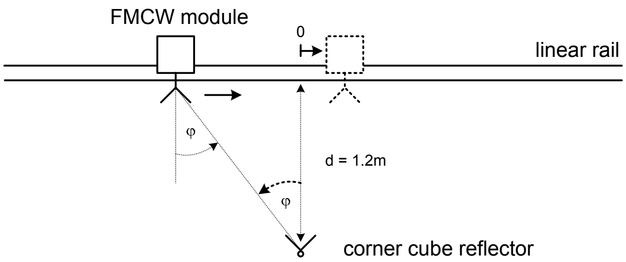

Fig. 4 Schematic of the measurement configuration. The FMCW radar is mounted on a rail and passes a static corner cube at a distance of 1.2 m.

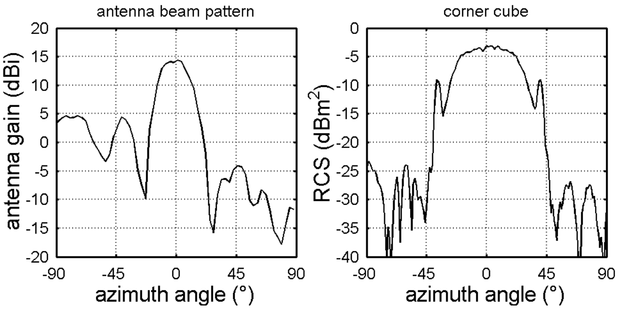

Fig. 5 Radiation pattern of the radar antennas (left), and the corner cube reflector (right), measured on a turn table in an anechoic chamber.

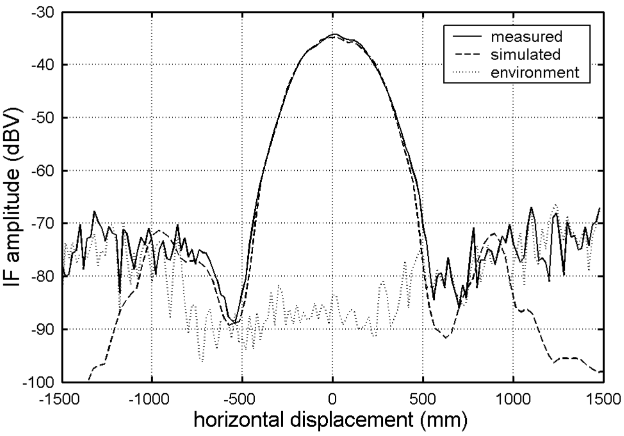

Fig. 6 Comparison of simulation and measurement of the pass-by scenario. Furthermore a reference measurement of the laboratory environment is shown.

The radiation patterns of both, the radar unit's antennas and the corner cube have been characterized beforehand in an anechoic chamber. The corresponding measurement results for the azimuth direction, which are subsequently loaded into the simulator, are shown in Fig. 5. Note that the asymmetric beam pattern of the antenna originates from absorber material, which is placed in between the antennas to improve the isolation. Fig. 6 compares the predicted and measured amplitude of the target's reflection with respect to the horizontal displacement of radar unit and corner cube. Simulation and measurement show very good agreement, especially in the center region. In the border areas, the un-modeled environmental reflections in the laboratory, which have been identified in a reference measurement without the target reflector, overlay the response from the corner cube.

IV. Conclusions

In this contribution, we presented a novel simulation environment for FMCW radar units. In contrast to commercially available hardware simulation tools, the new solution allows a flexible and joint treatment of hardware simulation and analysis of the influence of system control, such as the sweep parameters. From the output of the simulator valuable information for an optimization of the system's hardware as well as the control parameters can be drawn. Furthermore, due to the computationally efficient implementation, the generated data provides a powerful development and testing platform for signal processing algorithms. For existing systems an optimized interrogation strategy can be assessed.

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